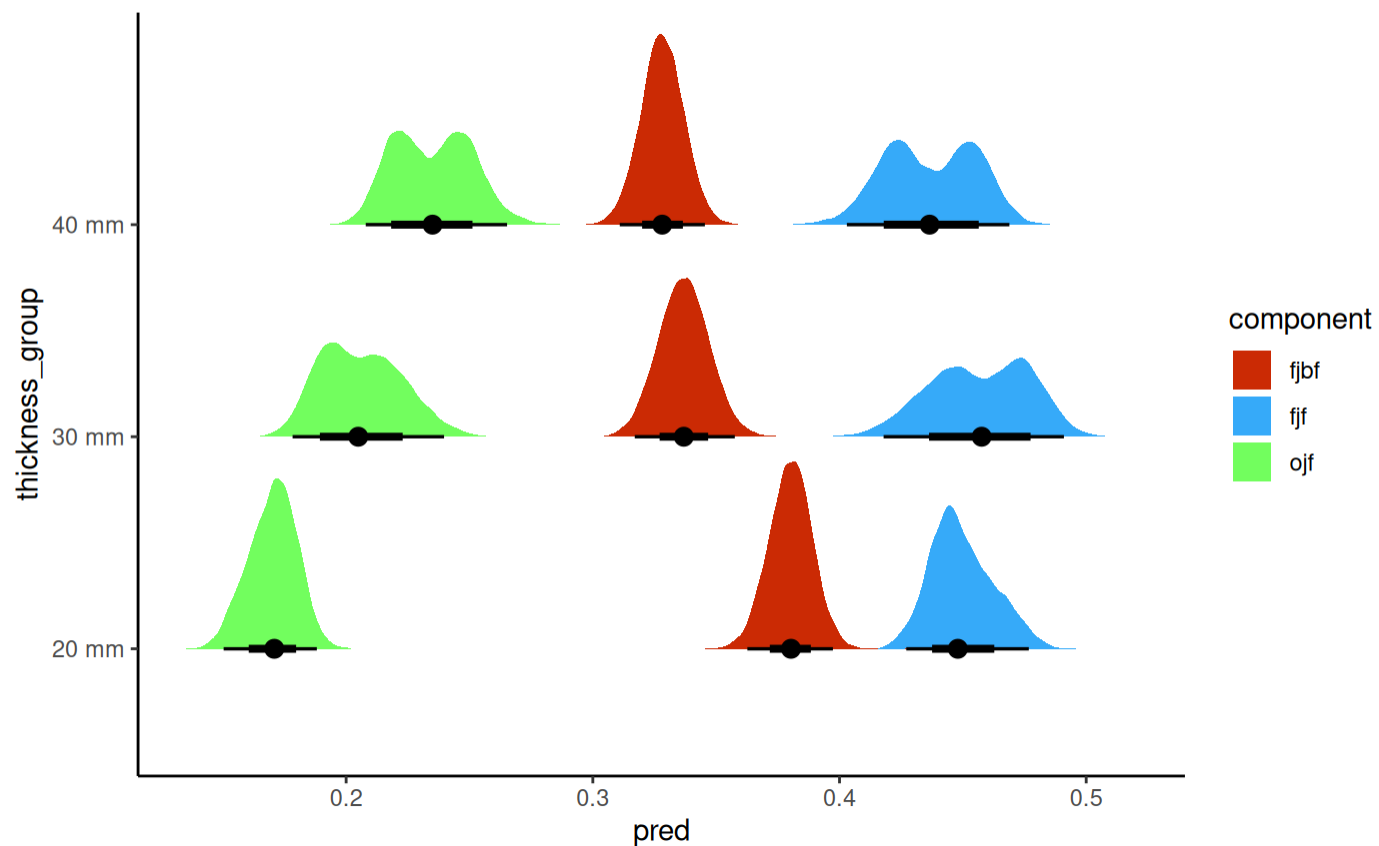


DSP I – Research data

Where does research data come from?



Where does research data come from?

- Experiments,
- Testing,
- Observation...

When do we collect research data?

When do we collect research data?

- When we have a research question or hypothesis
- Long-term monitoring
- Quality control testing

What do we do with a research question?

What do we do with a research question?

- Plan, execute, and analyse an experiment

Design & Analysis of Planned Experiments

What is an experiment?

What is an experiment?

- A study designed to provide data about a phenomenon
 - Attached to a very specific research question and hypothesis
- Controls nuisance factors
- Has specific structure (or study/experimental design)

What is study/experimental design?

- Specifies a methodology for data collection
- Where data is collected from
- Goals of analysis
- Some aspects of analysis are pre-defined

Three main components of experimental design

- Randomisation
- Replication
- noise Reduction

Randomisation

Randomisation allows us to generalise to broader populations than our sample

- A study investigating how effective blood pressure medicine is conducted on a sample of white males may not be application to women or non-white populations
- A study conducted on a sample of students from UP may not be generalisable to students from University of Helsinki, in Finland

Replication

Replication is distinct repetitions of an experiment

Very common for researchers to make mistakes with replication!!

Some related key concepts:

- Population: the broadest group you are studying
- Sample: the subset you include in your experiment
- Experimental units:
Subjects/Specimens... Individual units from within you specimen (persons, water samples, etc.)

Noise Reduction (nuisance control)

Eliminating or controlling for variables outside of your studies main interest.

Example: Testing batches of a coating system that are applied to many experimental units. Because it is prepared in batches, each batch is a supergroup of experimental units.

Common tools:

- *Blocking* subsets of your experiment repeated to control for confounding variables
- In statistics we use *blocks*, *split plots*, *whole plots* to account for this (terminology may vary...)
- We used mixed or hierarchal models to account for this in analysis. Special tools needed.

How do research questions and study designs influence analysis?

- Data structure (based on experimental design)
- Type of data collected (based research question and method of data collection)
- Both of these (↑) influence the type of analysis and the specific tools used.
- Hypothesis testing, causation / causal analysis

Example 1:

Research question: How do UP students view the use of AI in teaching?

Hypothesis: Students in computer science & mathematics view AI in teaching more favourably than students in other disciplines?

Experiment: Survey, using a custom questionnaire

Data required by hypothesis:

- *Student info:* degree programme
- *Opinion/view:* response to question: how do you view instructor use of AI to create teaching materials. Likert scale: very unfavourably, unfavourably, neutral, favourably, very favourably

Example 1:

What is the experimental unit?

What type of data is returned?

What other variables might we consider?

Example 1:

What is the experimental unit? *One student*

What type of data is returned? *Ordinal data*

What other variables might we consider? Age, year of study, gender, socioeconomic status, student use of AI/experience with AI

Example 2:

Research question: How does ambient noise affect student cognition?

Hypothesis: Students in favourable noise environments have improved cognitive performance

Experiment: Laboratory experiment

Data required by hypothesis:

- *Student info:* student status
- *Cognitive test result:* standard test (stroop, jack & jill)
- *Description of noise environment* (controlled, A B C)

Example 2:

What is the experimental unit?

What type of data is returned?

What other variables might we consider?

Example 2:

What is the experimental unit? *One student (tested once or multiple times)*

What type of data is returned? *Test score, bounded positive*

What other variables might we consider? Age, year of study, gender, socioeconomic status, student use of AI/experience with AI, preference for noise settings (music, quiet, nature sounds)

What does it mean for analysis?

Example 1:

The data from likert scales is **ordinal** – for responses like unfavourable -> favourable are converted to numbers (-2 to 2, for example) we don't know the spacing between those values. Cannot use tools like linear regression... ordered logistic regression

What does it mean for analysis?

Example 2:

Data are positive within a specific range, say 1 to 100. Normality may not be expected and linear regression can produce odd results – confidence intervals that exceed bounds. Linear regression may work, otherwise can use distributional regression (likely half-normal, gamma, or beta regression).

Within-subjects vs. between-subjects design